

Вариант 3;

1. Задаем начальные данные;

```
Subscript[x, 0] = 0; Subscript[x, 1] = Pi/4; Subscript[x, 2] = Pi/2; n = 2;
```

```
f[x_] = Sin[x]^2*Cos[x];
```

```
m = 3;
```

```
NN = (m + 1)*(n + 1) - 1;
```

2. Строим интерполяционный многочлен Эрмита;

```
P[x_] = Sum[Subscript[a, k] x^k, {k, 0, NN}];
```

```
eqv = Table[(D[P[x], {x, j}] /. x -> Subscript[x, k]) == (D[f[x], {x, j}] /. x -> Subscri
```

```
koef = Solve[eqv, {}] // Flatten;
```

```
P[x_] = P[x] /. koef // Expand // N[#, 16]&
```

```
 $x^2 - 0.8336508868486672 x^4 + 0.002308333158932346 x^5 +$ 
```

```
 $0.2457053828758706 x^6 + 0.01173278737980559 x^7 - 0.05178294615415484 x^8 +$ 
```

```
 $0.005494713478560081 x^9 + 0.003446201415854223 x^{10} - 0.0006799256305951446 x^{11}$ 
```

3. Проверяем выполнение интерполяционных условий;

```
Table[(D[P[x], {x, j}] /. x -> Subscript[x, k]) - (D[f[x], {x, j}] /. x -> Subscript[x,
```

```
{True, True, True}, {True, True, True}, {True, True, True}, {True, True, True}]
```

4. Контроль через InterpolatingPolynomial;

```
Tbl = Table[{Subscript[x, i], Table[D[f[x], {x, j}] /. x -> Subscript[x, i], {j, 0, m}]},
```

```
P1[x_] = InterpolatingPolynomial[Tbl, x] // Expand;
```

```
N[P[x] - P1[x]] // Chop
```

```
0
```

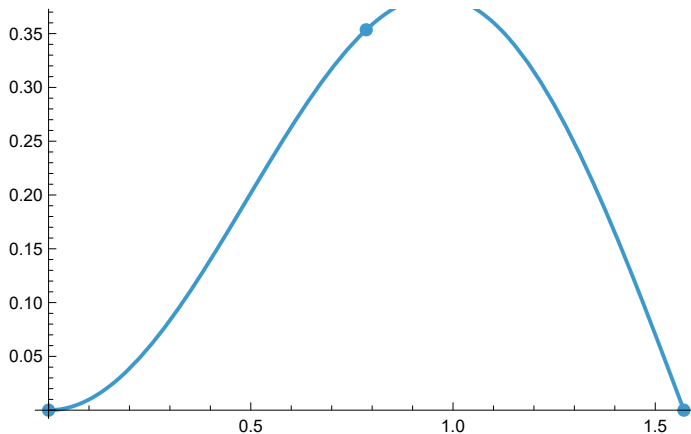
5. Изображаем точки и многочлен;

```
Tb = Table[{Subscript[x, i], f[Subscript[x, i]]}, {i, 0, n}];
```

```
Gr1 = ListPlot[Tb, PlotStyle -> {PointSize[0.02]}];
```

```
Gr2 = Plot[P[x], {x, Subscript[x, 0], Subscript[x, n]}];
```

```
Show[Gr1, Gr2]
```



6. Оценка погрешности;

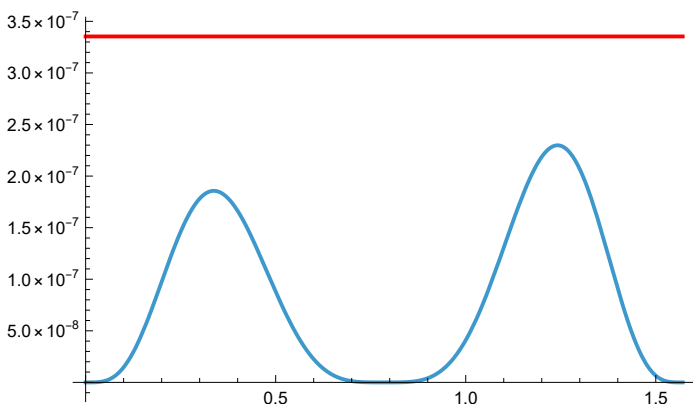
```
M1 = Maximize[{D[f[x], {x, NN + 1}], Subscript[x, 0] <= x <= Subscript[x, n]}, x][[1]];
M2 = Minimize[{D[f[x], {x, NN + 1}], Subscript[x, 0] <= x <= Subscript[x, n]}, x][[1]];
M = Max[{Abs[M1], Abs[M2]}];
omega[t_] = Product[(t - Subscript[x, i])^(m + 1), {i, 0, n}];
Momega = Maximize[{Abs[omega[x]], Subscript[x, 0] <= x <= Subscript[x, n]}, x][[1]];
R = M/((NN + 1)!) * Momega // N
Print["Ответ: оценка погрешности R = ", R]
```

Ответ: оценка погрешности $R = 3.35374 \times 10^{-7}$

3.35374×10^{-7}

7. Графики абсолютной погрешности и оценки;

```
Gr3 = Plot[Abs[f[x] - P[x]], {x, Subscript[x, 0], Subscript[x, n]}];
Gr4 = Plot[R, {x, Subscript[x, 0], Subscript[x, n]}, PlotStyle -> Red];
Show[Gr3, Gr4, PlotRange -> Automatic]
```



8. Убеждаемся что погрешность не превосходит оценки;

```
MR1 = Maximize[{f[x] - P[x], Subscript[x, 0] <= x <= Subscript[x, n]}, x][[1]];
MR2 = Minimize[{f[x] - P[x], Subscript[x, 0] <= x <= Subscript[x, n]}, x][[1]];
MR = Max[{Abs[MR1], Abs[MR2]}];
MR <= R
```

True

9. Проверяем свойство инвариантности;

```
Unprotect[Power];
0^0 := 1;
Table[Module[{g, Ht}, g[t_] = t^jj; Ht[t_] = InterpolatingPolynomial[Table[{Subscript[x, i]
{{1, True}, {t, True}, {t^2, True}, {t^3, True}, {t^4, True}, {t^5, True}, {t^6, True},
{t^7, True}, {t^8, True}, {t^9, True}, {t^10, True}, {t^11, True}, {t^12, False}}
```